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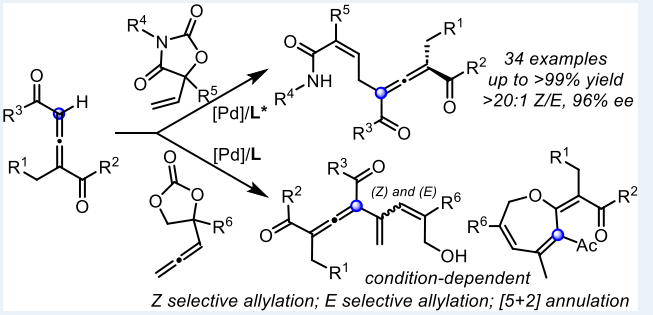
Palladium-Catalyzed Allylic Alkylation Reactions of Nucleophilic Allenones: Asymmetric Allylic Alkylation, *Z/E* Divergent Allylic Alkylation, and [5 + 2] Annulation

Wang Wang,[‖] Yimin Hu,[‖] Kuan Li, Jiaqing Xu, Cheng Zhang, Leijie Zhou,* Lihan Zhu,* and Hongchao Guo*

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ABSTRACT: Metal-catalyzed allenylic substitution reactions where allenes serve as electrophilic precursors have been recognized as a rapid way for novel allene construction. On the contrary, the chemistry in which allenes act as nucleophiles has been far less investigated, especially in a powerful platform such as metal-catalyzed allylic alkylation reactions. We herein describe two unprecedented palladium-catalyzed allylic alkylation reactions of an allene nucleophile. In the first reaction, using vinyloxazolidinones as the allyl precursor, the palladium-catalyzed asymmetric allylation of trisubstituted allenones worked well to prepare an array of axially chiral tetrasubstituted allenes. Mechanistic studies and density functional theory (DFT) calculations indicated that weak hydrogen-bonding interaction between the acidic C(sp^2)−H of allenone and the nitrogen anion of chiral π-azaallyl-Pd species is key to the success of stereocontrol. This reaction revealed the intriguing reactivity of nucleophilic allenones in a metal-catalyzed asymmetric allylation reaction for the first time. In the second reaction, with the use of allenylethylene carbonates as π-oxyallyl-Pd precursors, the palladium-catalyzed allylic alkylation reaction of allenone presented unique reactivity under different reaction conditions to provide divergent synthetic access to (*E*)- and (*Z*)-allenyl diene products. Interestingly, the subsequent sequential intramolecular cyclization/isomerization reaction of the (*Z*)-product delivered dihydrooxepine derivatives as [5 + 2] annulation products.



KEYWORDS: *asymmetric catalysis, palladium catalysis, allylation, annulation, chiral allene, axial chirality*

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INTRODUCTION

Allene motifs ubiquitously existed in natural products, bioactive molecules, and functional materials.[1] Therefore, the development of catalytic systems that allow the synthesis of various allenes, especially the challenging multisubstituted allenes, has attracted a lot of attention from chemical communities. This research topic has been greatly expanded during the past two decades, and diverse elegant strategies have been established for the efficient synthesis of allenes. Among them, strategies via catalytic functionalization of achiral or racemic substrates for direct construction of functionalized allenes in both enantio- and nonenantioselective patterns have drawn particular interest.[2−4] In this context, the research group of Ma is listed as one of the main contributors. Various enantioselective catalytic protocols such as β-elimination, allenylic and propargylic substitution, Heck reaction, Suzuki coupling, alkynyl ketimine addition, and allenylation of alkynes with aldehydes have been developed by the group, allowing convenient synthesis of structurally diverse functionalized allenes.[4] Recently, endeavors have been made to prepare functionalized allenes by using the Pd- or Ir-catalyzed allenylic substitution protocols, where racemic or achiral allenes served as electrophiles to couple with diverse nucleophiles.[5] As a result, a variety of glorious designs including mono- and dual catalysis have been demonstrated, making this strategy one of the most straightforward and reliable pathways to access functionalized allenes in both enantio- and nonenantioselective patterns (Scheme 1a).[2−5] However, it was usually mandatory for such a strategy to have specific preinstalled leaving groups in the allenylic position to form the key π-methyleneallylic palladium intermediate. This limited the types of both allene substrates and corresponding allene products to a certain extent. Thus, developing alternative catalytic protocols that allow the use of expanded types of allenes would be of great

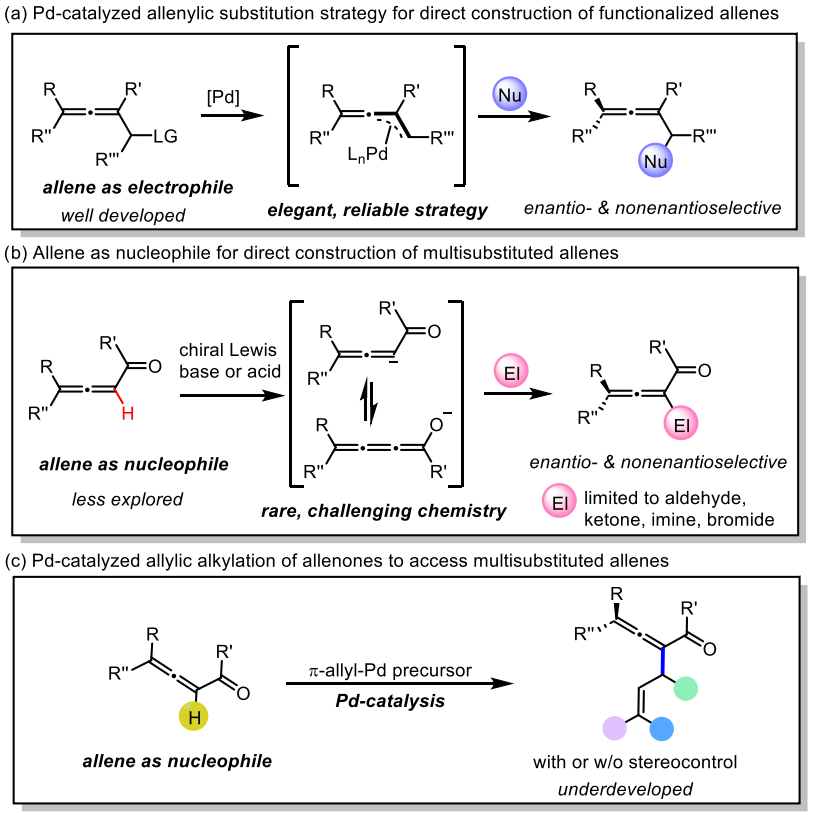
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Scheme 1. Catalytic Strategies with Allenes as Electrophiles and Nucleophiles for the Synthesis of Functionalized Allenes



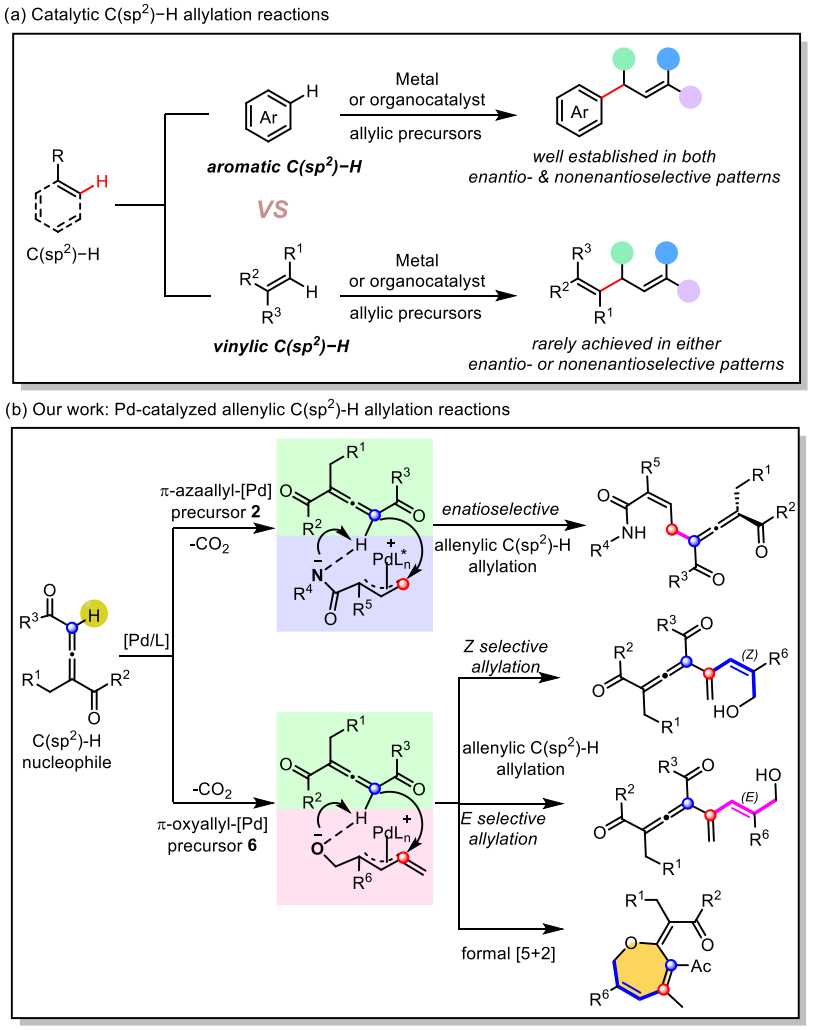
significance for the diverse synthesis of novel multisubstituted allene compounds.

In 1993, the Tsuboi group reported an aldol condensation of nucleophilic allenes with aldehydes, revealing the fascinating potential of allenes to act as nucleophiles.[6a] In 2009, Miller and co-workers described the first enantioselective addition of allenoate to N-acyl imines under the catalysis of a chiral peptide catalyst.[6b] Within both works, electron-deficient allenes bearing a C(sp^2)−H bond activated in situ serve as precursors for the cumulenolate anion or its equivalent, which could be trapped by various electrophiles in an enantio- or nonenantioselective way (Scheme 1b). Indeed, this idea was further verified soon by later works of Miller,[6c] Maruoka,[7] Feng/Liu,[8] Yu,[9] and List.[10] To date, electrophiles such as aldehydes, ketones, and aryl and allylic bromides have been successfully employed via asymmetric Lewis base or acid catalysis.[6−10] However, the state of the art is still limited to the previously mentioned several examples. Novel catalytic coupling processes of the nucleophilic cumulenolate anion with various types of electrophiles are still in high demand. In 2020, exploiting this cumulenolate anion chemistry, our group disclosed asymmetric allylic alkylation of allenones and Morita−Baylis−Hillman carbonates under organocatalysis.[11] Nevertheless, the catalytic method for the coupling of the cumulenolate anion with widely used π-allyl-Pd intermediates in either the enantio- or nonenantioselective pattern is still unknown to date (Scheme 1c).

On the other hand, the direct allylation reactions of the C(sp^2)−H bond have attracted broad attention due to the ubiquitous presence of allyl aromatic rings and allyl olefins in bioactive molecules.[12] Currently, the direct allylation reactions of the aromatic C(sp^2)−H bond have been thoroughly studied via metal catalysis, organocatalysis, or dual catalysis in both asymmetric and nonasymmetric patterns.[13,14] In comparison, the direct allylation of the vinylic C(sp^2)−H bond, especially in an enantioselective way, was less investigated because of a lack of viable activation mode (Scheme 2a). To date, successful examples of direct allylation of the vinylic C(sp^2)−H bond were very rare.[15] In this regard, as mentioned above, we recently disclosed direct allylation of the allenylic C(sp^2)−H bond via organocatalysis.[11] Considering that allylpalladium intermediates were extremely versatile in asymmetric allylation reactions of the C(sp^3)−H bond,[14] we envisioned that the allylation of the allenylic C(sp^2)−H bond through palladium catalysis could be feasible. Our laboratory has been devoted to developing diverse annulation and allylation reactions with the use of palladium catalysis, focusing on the development of new precursors of allylpalladium intermediates.[16] In 2021, we synthesized allenylethylene carbonates (AECs) as highly reactive precursors of π-oxyallyl-Pd species, which displayed good performance in several interesting transformations.[16a] Very recently, we reported a fungicide-inspired precursor of π-allylpalladium intermediates that underwent [5 + 3] and [3 + 2] cycloadditions as well as allylation reaction.[16e] Combining our continuous interest in both the allenylic C(sp^2)−H nucleophile and the diverse reactivity of the novel precursors of π-allylpalladium intermediates, we herein present our results on allylic alkylation reactions of nucleophilic allenones. In the first part, enantioselective coupling of the allenones with π-azaallyl-Pd species gave rise to axially chiral allylic allene products, which represents the first example of the Pd-catalyzed asymmetric allylation reaction of the allenylic C(sp^2)−H nucleophile. In the other part, we realized Pd-catalyzed *Z*/*E* divergent allylic alkylation reactions of

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Scheme 2. Catalytic C(sp^2)−H Allylation Reactions



nucleophilic allenones and the π-oxyallyl-Pd precursor as well as an interesting [5 + 2] annulation via the sequential intramolecular cyclization/isomerization reaction of the (*Z*)-allylic product (Scheme 2b).

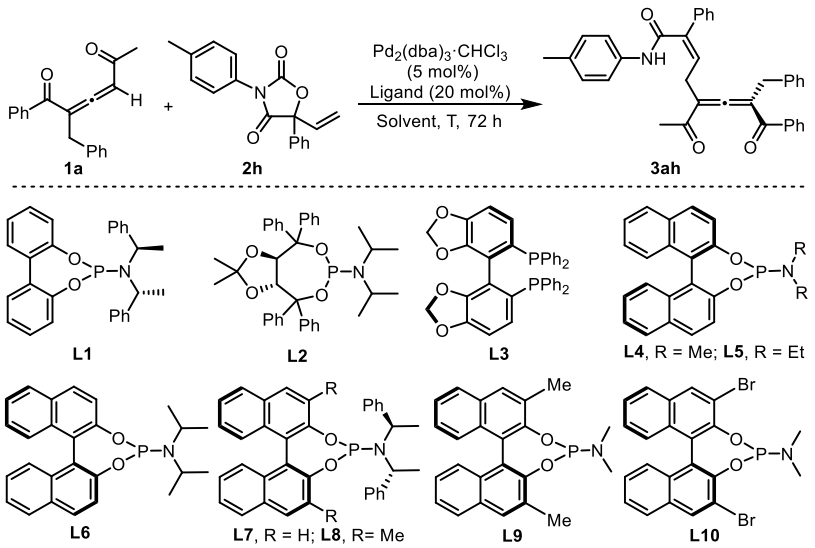
RESULTS AND DISCUSSION

We initiated our study on the asymmetric allylic alkylation reaction by using trisubstituted allenone **1a** and 5-vinyl-oxazolidine-2,4-dione **2h** as model substrates in the presence of 5 mol % $Pd_2(dba)_3{\cdot}CHCl_3$ and 20 mol % ligands at 25 °C (Table 1). At first, employment of biphenol and TADDOL-based phosphoramidites (**L1**−**L2**) and Segphos (**L3**) as the ligand failed to give the desired product (entries 1−3). Switching to the typical BINOL-based phosphoramidite ligand **L4** successfully produced the desired allylation product **3ah** in 60% yield, albeit with a low ee and a moderate *Z*/*E* ratio (entry 4). Variation of the dimethyl amine group of **L4** largely suppressed or shut down the reaction (entries 5−7, **L5**−**L7**). It was observed that the use of ligand **L8** significantly improved the enantioselectivity and the *Z*/*E* selectivity of the reaction (entry 8). We were happy to find that when 3,3′-dimethyl-substituted ligand **L9** was used, the reaction proceeded smoothly to afford product **3ah** in 84% yield with 69% ee and 9:1 *Z*/*E* (entry 9), whereas 3,3′-dibromo-substituted ligand **L10** displayed poor reactivity and *Z*/*E* selectivity (entry 10). To our delight, decreasing the reaction temperature to 0 °C greatly increased the ee value of product **3ah** to 89%, albeit in slightly decreased yield (62% yield, entry 11). We then evaluated the solvent effect at 0 °C. It was found that when THF was employed as the solvent, higher ee and *Z*/*E* ratio values of **3ah** were obtained (entry 12, 63% yield, 94% ee, > 20:1 *Z*/*E*). Other solvents such as toluene, Et_2O, MTBE, dioxane, and $CHCl_3$ failed to deliver better results (entries 13−17). Because of the low solubility of substrates in Et_2O, the reaction did not work (entry 14). When dioxane was used as the solvent, ee sharply dropped to 17%, which was because the strong polarity of dioxane disturbed the effective interaction between the substrate and the chiral catalyst (entry 16). In addition, the ratio of [Pd]/ligand was also studied in the reaction. Decreasing the loading of **L9** from 20 to 15 mol % led to remarkably deteriorated yield (entry 18), while 10 mol % loading of **L9** caused a complete loss of reactivity (entry 19).

With the optimized reaction conditions in hand, we first explored the substrate scope of trisubstituted allenones **1** (Scheme 3). In general, allenones with electron-withdrawing or -donating groups on the phenyl ring of the benzyl and benzoyl group all performed the reaction well, delivering the corresponding products in good to excellent *Z*/*E* selectivities and moderate to high yields and enantioselectivities (**3bh**−**3nh**). Allenones bearing various *meta*- or *para*-substituted

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Table 1. Optimization of the Asymmetric Allylic Alkylation Reaction[a]



entry	ligand	solvent	yield (%)[b]	Z/E[c]	ee (%)[d]
1	**L1**	CH_2Cl_2	trace		
2	**L2**	CH_2Cl_2	trace		
3	**L3**	CH_2Cl_2	trace		
4	**L4**	CH_2Cl_2	60	4:1	26
5	**L5**	CH_2Cl_2	27	2.5:1	16
6	**L6**	CH_2Cl_2	trace		
7	**L7**	CH_2Cl_2	trace		
8	**L8**	CH_2Cl_2	48	>20:1	66
9	**L9**	CH_2Cl_2	84	9:1	69
10	**L10**	CH_2Cl_2	34	2:1	51
11[e,h]	**L9**	CH_2Cl_2	62	9:1	89
12[e,h]	**L9**	**THF**	**63**	>20:1	**94**
13[e]	**L9**	toluene	67	3:1	42
14[e]	**L9**	Et_2O	trace		
15[e]	**L9**	MTBE	61	10:1	88
16[e]	**L9**	dioxane	90	4:1	17
17[e]	**L9**	$CHCl_3$	84	5:1	91
18[e,f]	**L9**	THF	38	10:1	91
19[e,g]	**L9**	THF	trace		

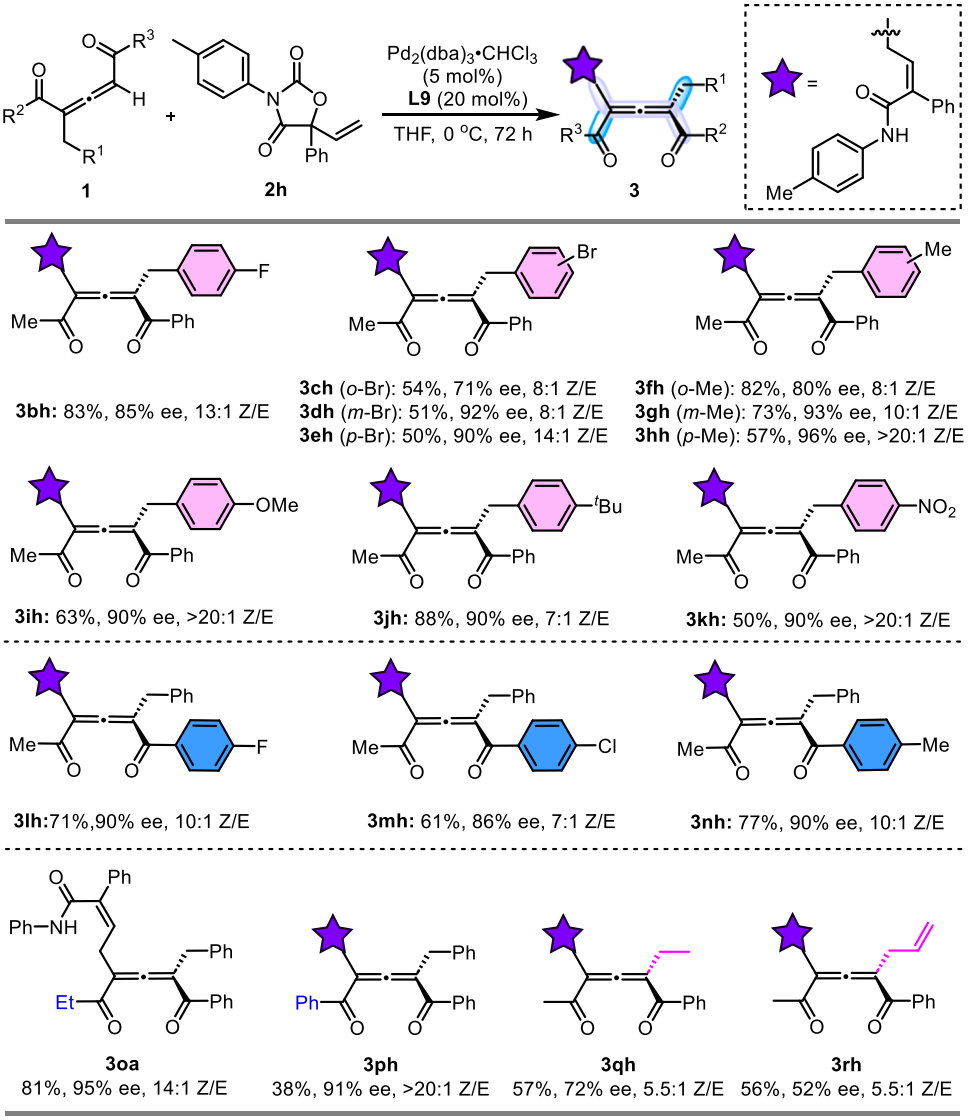
[a]Unless otherwise indicated, all reactions were performed with **1a** (0.05 mmol), **2h** (0.06 mmol) in 1 mL of solvent at 25 °C for 72 h in the presence of $Pd_2(dba)_3$·$CHCl_3$ (5 mol %), and a ligand (20 mol %). [b]Isolated yield. [c]Determined by ^{1}H NMR analysis. [d]Determined by chiral HPLC analysis. [e]The reaction was performed at 0 °C. [f]**L9** (15 mol %) was used. [g]**L9** (10 mol %) was used. [h]The reaction was performed on a 0.10 mmol scale with respect to **1a** in 1 mL of solvent.

phenyl groups on the R^1 position were converted into the corresponding products **3bh**−**3kh** in moderate to high yields with excellent stereoselectivity (50−88% yields, up to 20:1 Z/E, 71−96% ee), while allenones with a strong electron-withdrawing group at the R^1 position showed relatively lower reactivity (**3ch**−**3eh**, **3kh**, 50−54% yields). The allenones having an *ortho*-substituted R^1 group were also suitable substrates for the reaction, albeit presenting moderate stereoselectivity (**3ch**, **3fh**). Benzoyl allenones with F-, Cl-, and Me-substituted phenyl rings were well tolerated in the reaction, too, affording desired products with satisfactory results. Furthermore, we were pleased to find that replacing the methyl group at R^3 with an ethyl group enhanced reactivity, enantioselectivity, and Z/E selectivity (**3oa**, 81% yield with 14:1 Z/E and 95% ee). To our delight, besides allenones having alkyl as R^3, allenone **1p** with phenyl as R^3 also performed the reaction well, providing **3ph** in excellent enantioselectivity and Z/E selectivity, albeit with diminished yield (38% yield, >20:1 Z/E, 91% ee). Finally, we investigated the influence of variations on R^1. It was found that allenone **1q** having methyl as R^1 underwent the reaction smoothly to give product **3qh** in 57% yield with 5.5:1 Z/E and 72% ee. When R^1 was a vinyl group (**3rh**), moderate reactivity and selectivity was observed. These results indicated that the aromatic group at R^1 of allenone is critical to ensure high reactivity and excellent stereocontrol during the reaction process. We also investigated the influence of the ketone group. It was found that replacing this ketone group with the Bn or ester group completely prevented the reaction from occurring (see the Supporting Information (SI) for more information).

Next, we investigated the scope of 5-vinyloxazolidine-2,4-diones **2** (Scheme 4). 5-vinyloxazolidine-2,4-diones **2** bearing various substituted aromatic (R^4) groups all worked well in the current reaction, regardless of *meta*- or *para*-substitution or the

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Scheme 3. Scope of Trisubstituted Allenones 1 for the Asymmetric Allylic Alkylation Reaction[a]



[a]Unless otherwise indicated, all reactions were performed with **1** (0.1 mmol), **2h** (0.12 mmol) in 1 mL of THF at 0 °C for 72 h in the presence of $Pd_2(dba)_3$·$CHCl_3$ (5 mol %), and a ligand (20 mol %); all yields are isolated yields; *Z*/*E* ratios were determined by ^{1}H NMR analysis; ee values were determined by chiral HPLC analysis; **2a** was used instead of **2h** in the reaction of product **3oa**.

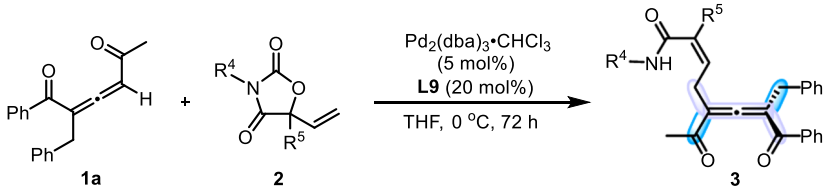
electron properties of the substituents (**3aa**−**3am**, 52−97% yields, 83−94% ee and up to >20:1 *Z*/*E*). Interestingly, compared with its *meta*-OMe counterpart **2i**, substrate **2j** bearing a *para*-OMe-substituted phenyl ring presented an overall decrease of both reactivity and stereoselectivity, which may be due to the fact that the hydrogen-bonding ability of the nitrogen ion intermediate generated from *pare*-substituted **2j** was slightly weaker. Besides, variations of α-substituted (R^5) substrates **2** were also explored (**3an**−**3av**). The F-, Cl-, Br-, and Me-substituted substrates all displayed satisfactory reactivity and stereoselectivity in this reaction, affording axially chiral allylic allene products in 87−99% yields, along with 81−93% ee and 7:1−10:1 *Z*/*E* ratios (**3an**−**3at**). Unfortunately, methoxy-substituted **2u** and **2v** were not well compatible with the current system, producing products **3au** and **3av** in lower 37 and 42% yields, respectively, which may be because the increased electron density of the aromatic motif reduced the reactivity of the allylpalladium intermediate. Additionally, the absolute configuration of the products was unambiguously assigned through the X-ray crystallographic analysis of product **3aq**.[17]

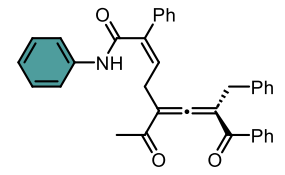
To demonstrate the application potential of this reaction, a scale-up reaction was conducted (Scheme 5a), affording the desired product **3aq** in 96% yield (538 mg), along with 90% ee and 7:1 *Z*/*E*. The axially chiral allene product **3aq** was found to be highly reactive. It furnished a thermal [3 + 2] cycloaddition with C,N-cyclic azomethine imine **4** in chloroform at 90 °C, delivering chiral hexahydropyrazole[5,1-*a*]isoquinoline derivatives **5** in 68% yield with >20:1 dr and 76% ee (Scheme 5b). The accurate structure of product **5** has been unambiguously confirmed through X-ray crystallographic analysis.[17]

To unveil the reaction mechanism, the following experiments were conducted (Figure 1). First, racemic substrates **1a** and **2q** were subjected to the reaction under standard reaction conditions, and the yield and ee of product **3aq** and recovered substrate **2q** were monitored for 48 h. It was observed that a high ee of **3aq** was detected at the beginning and remained

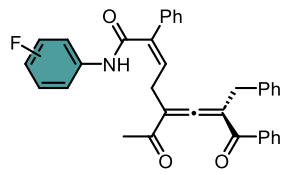
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Scheme 4. Scope of 5-Vinyloxazolidine-2,4-diones 2 for the Asymmetric Allylic Alkylation Reaction[a]

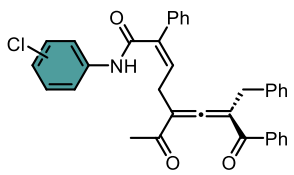




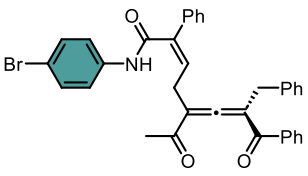
3aa: 70%, 93% ee, 11:1 Z/E



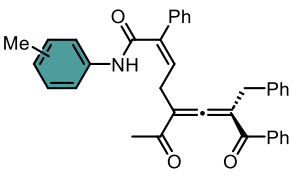
3ab (*m*-F): 67%, 92% ee, 14:1 Z/E
3ac (*p*-F): 90%, 86% ee, 12.5:1 Z/E



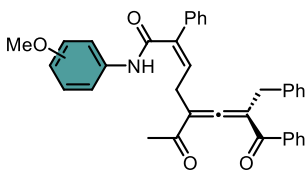
3ad (*m*-Cl): 82%, 94% ee, 14:1 Z/E
3ae (*p*-Cl): 90%, 90% ee, 14:1 Z/E



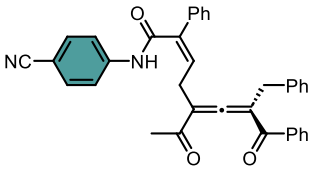
3af: 86%, 85% ee, >20:1 Z/E



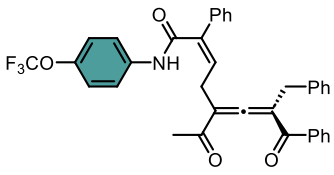
3ag (*m*-Me): 67%, 90% ee, 10:1 Z/E
3ah (*p*-Me): 63%, 94% ee, >20:1 Z/E



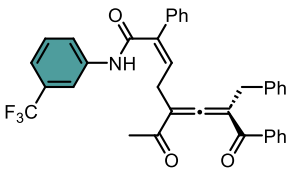
3ai (*m*-OMe): 86%, 90% ee, >20:1 Z/E
3aj (*p*-OMe): 52%, 90% ee, 11:1 Z/E



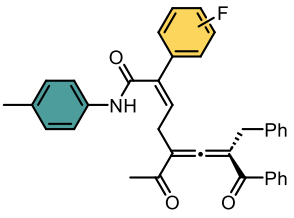
3ak: 85%, 90% ee, 11:1 Z/E



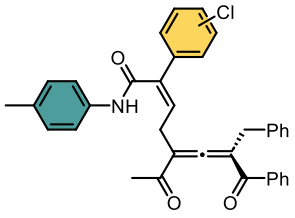
3al: 80%, 86% ee, 11:1 Z/E



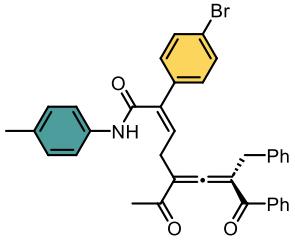
3am: 97%, 83% ee, 12.5:1 Z/E



3an (*m*-F): 99%, 91% ee, 10:1 Z/E
3ao (*p*-F): 99%, 90% ee, 8:1 Z/E

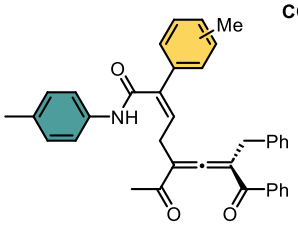


3ap (*m*-Cl): 99%, 81% ee, 10:1 Z/E
3aq (*p*-Cl): 98%, 93% ee, 7:1 Z/E

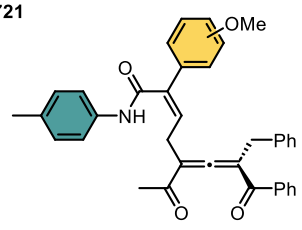


3ar: 92%, 90% ee, 7:1 Z/E

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3as (*m*-Me): 88%, 90% ee, 10:1 Z/E
3at (*p*-Me): 87%, 91% ee, 10:1 Z/E



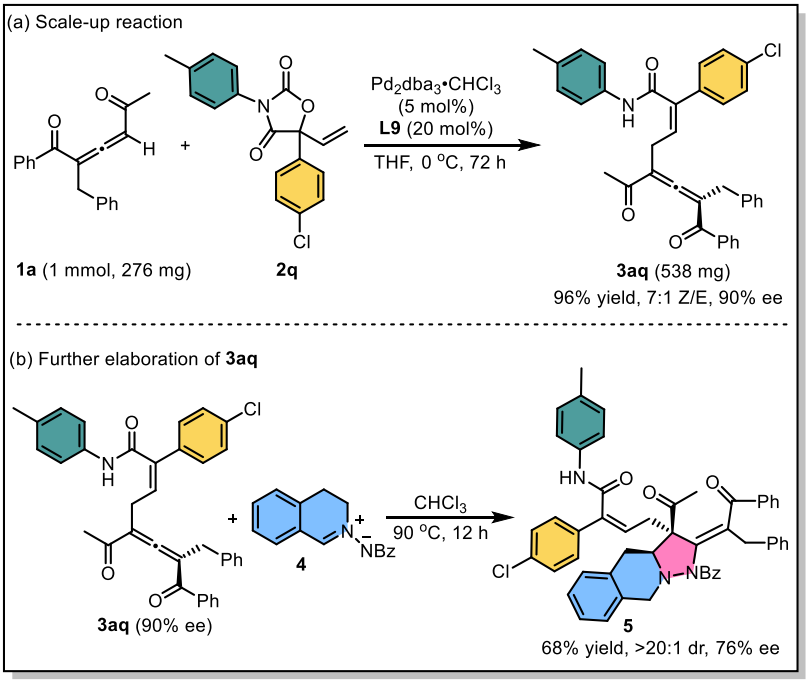
3au (*m*-OMe): 37%, 96% ee, 7.5:1 Z/E
3av (*p*-OMe): 42%, 84% ee, 11:1 Z/E

[a]Unless otherwise indicated, all reactions were performed with **1a** (0.1 mmol), **2** (0.12 mmol) in 1 mL of THF at 0 °C for 72 h in the presence of $Pd_2(dba)_3{\cdot}CHCl_3$ (5 mol %), and a ligand (20 mol %); all yields are isolated yields; *Z*/*E* ratios were determined by ^1H NMR analysis; ee values were determined by chiral HPLC analysis.

stable until the reaction finished (Figure 1a, I). At the same time, the ee of recovered **2q** continued to increase as the reaction proceeded over the course of 48 h. According to the results, both (*R*)- and (*S*)-**2q** were consumed, but (*R*)-**2q** was consumed quickly and (*S*)-**2q** was consumed slowly through racemization; therefore, (*S*)-**2q** was left to be recovered. These results suggested that the reaction underwent a weak dynamic kinetic resolution (DKR) process (Figure 1a, II). Subse-

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Scheme 5. Scale-up Reaction and Derivatization of Product 3aq



quently, the relationship between the ee of **L9** and product **3aq** was studied over 72 h of reaction time. The result indicated a nonlinear relationship between the ligand and the product, proving that the reactive chiral Pd catalyst was formed through two molecules of ligands coordinating with one palladium atom (Figure 1a, III). As shown in Figure 1b, several more experiments were conducted to gain a further understanding of the DKR process. The reaction of *rac*-**1a** and *rac*-**2q** with **L9** as a chiral ligand under the standard reaction conditions afforded product **3aq** in 98% yield with 93% ee. However, when *ent*-**L9** was used as the ligand, the corresponding product *ent*-**3aq** was produced with a slight decrease of yield and ee (Figure 1b, left, 94% yield and 89% ee). When *rac*-**1a** was treated with (*S*)-**2q** (82% ee) under the standard reaction conditions, product **3aq** was produced with 32% yield and 89% ee, while (*S*)-**2q** was recovered in 70% yield with 97% ee, indicating that (*R*)-**2q** was a more reactive enantiomer than (*S*)-**2q**. Replacing chiral ligand **L9** with *ent*-**L9** resulted in *ent*-**3aq** as the product. At the same time, (*S*)-**2q** was recovered in decreased ee (Figure 1b, right).

As shown in Figure 2, DFT calculations were carried out with the Gaussian 09 software package with the purpose of gaining a comprehensive understanding of the reaction mechanism between trisubstituted allenone (**1a**) and 5-vinyloxazolidine-2,4-dione (**2a**) under the catalysis of **L9**-Pd. Our calculations indicated that the reaction started from the olefin C1═C2 of the **2a**-coordinated Pd(0) complex, which was set to the relative zero point. The transition state for C3−O1 bond cleavage via **TS1** had an energy barrier of 18.5 kcal/mol to afford Pd(II)-π-allyl complex **int1**. Then, the second CO_2 elimination process occurred and the corresponding energy barrier was 24.5 kcal/mol (via **TS2**) to form a stable intermediate (**int2**). This step was highly exothermic by 28.8 kcal/mol. Substrate **2a** was added to the system through a C−H···N^- hydrogen bond interaction with the release of CO_2 to give **int3**, which was 8.4 kcal/mol higher than **int2**. Subsequently, the activation barrier of the transition state **TS3** for H1 shifting from the C5 of **2a** to N1 was 23.1 kcal/mol, and carbanion C5 was generated in **int4**. Cumulenolate intermediate **int4′** was also considered in the energy profile diagram, which was 8.5 kcal/mol lower than the energies of **int4** (see the SI for detailed information). The enantioselective C−C bond formation step **TS4_re_Z** (22.0 kcal/mol corresponds to **int3**) led to **3aa**, which involved a nucleophilic attack to the C1 of Pd-π-allyl by carbanion C5. The final C−C bond formation process was both the rate-determining and enantioselective step.

To elucidate the origin of the *Z*/*E* selectivities and enantioselectivities for **3aa**, we further located the **TS4_*re*_E**, **TS4_*si*_Z**, and **TS4_*si*_E** structures. Our calculations revealed that the **TS4_*re*_E** and **TS4_*si*_Z** transition states, which form the enantiomeric ***re*-E** and ***si*-Z** products, are 1.4 and 1.9 kcal/mol higher in energy than **TS4_*re*_Z**, leading to 13:1 *Z*/*E* ratios and 94% ee, in accord with our experimental findings (11:1 *Z*/*E* ratios and 93% ee, Table 2).

To evaluate the origin of stereoselectivities shown in Table 2, we employed the AIM analysis to identify important noncovalent interactions in these four transition states,[18] and the strengths of various hydrogen-bonding interactions were quantified using the method described by Espinosa et al. (Table S11−S13 in the SI).[19] The hydrogen bonding in **TS4_*re*_Z** features the largest number of weak interactions, including C−H···O (A1−A4) and C−H···π (A5−A7) between the BINOL skeleton of the ligand and the phenyl ring and carbonyl moiety of substrates **1a** and **2a**, which was estimated to be −15.4 kcal/mol in total. Similarly, such C−H···O and C−H···π hydrogen-bonding interactions also exist in the structure of **TS4_*re*_E** (B2−B5), which would provide the E product with the R configuration. Although addition of N−H···O (B1) could also contribute to the stability of **TS4_*re*_E** by −8.2 kcal/mol, the total bond energy of the hydrogen bonding in **TS4_*re*_E** was estimated to be − 14.5 kcal/mol, which is around 0.9 kcal/mol weaker than the one in **TS4_*re*_Z**, thus accounting for the preference of **TS4_*re*_Z** over **TS4_*re*_E** by

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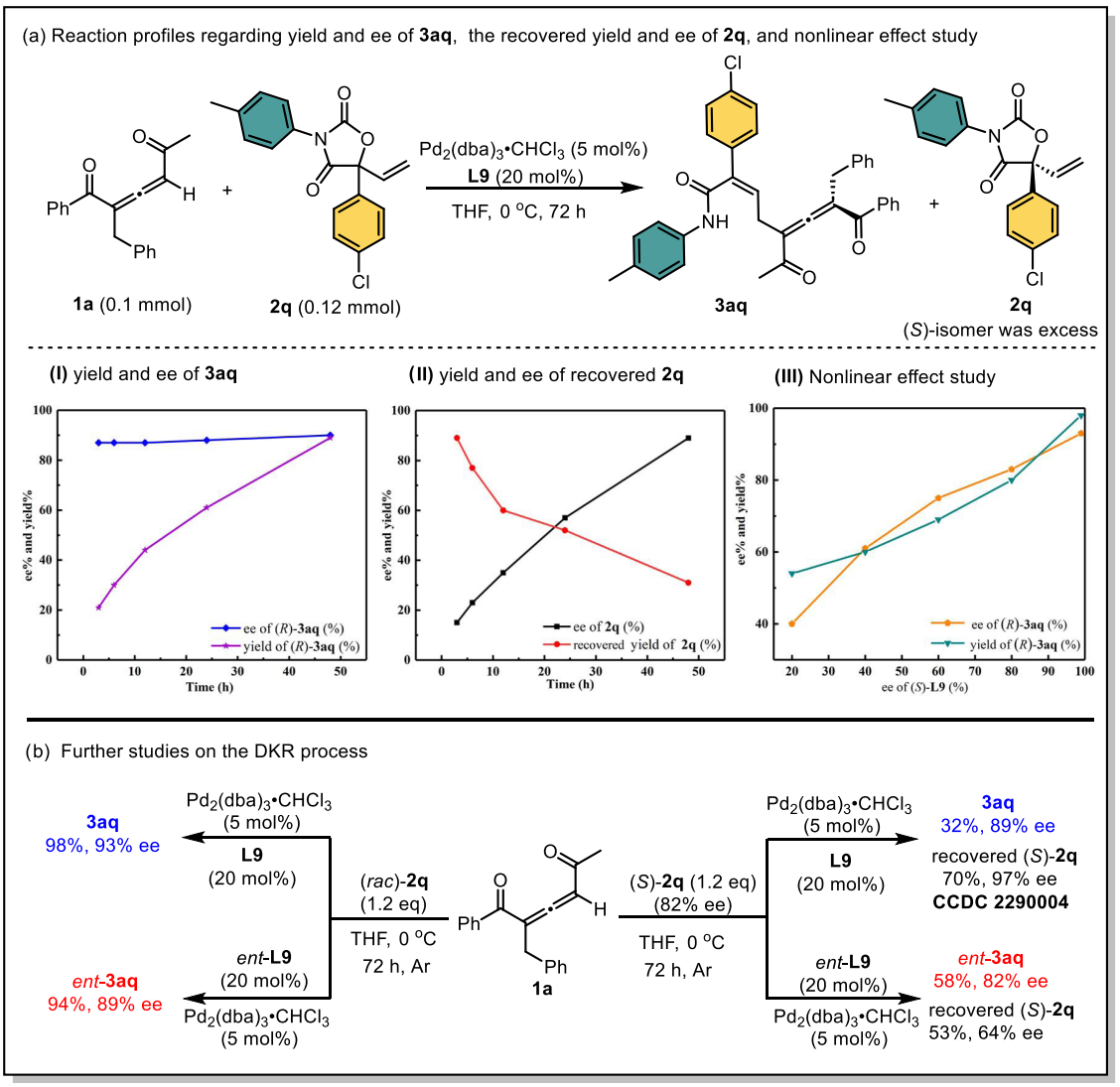


Figure 1. Studies on the reaction mechanism of the asymmetric allylic alkylation reaction.

1.4 kcal/mol. Furthermore, **TS4_*si*_Z** is also stabilized by the hydrogen bonding formed between the ligand's C−H and bridged oxygen, the carbonyl oxygen of **1a** and **2a**, and the C−H and π of the phenyl of substrate **1a**. The bond energy of the hydrogen bonding in **TS4_*si*_Z** was estimated to be about −13.2 kcal/mol. Therefore, the different strength of noncovalent interactions leads to the lower stability of **TS4_*si*_Z** than **TS4_*re*_Z** by around 1.9 kcal/mol. In addition, the hydrogen bonding existing in **TS4_*si*_E** decreases to four, named D1−D4, single N−H···O, double C−H···O, and single C−H···π interactions. In the transition state leading to the **E** product with S configuration, the phenyl ring of substrate **1a** orients to the vacancy of the catalyst. This could not provide the C−H···O interactions as in **TS4_*re*_Z** (A1) and **TS4_*si*_Z** (C2) and also lost the C−H···π interactions as shown in **TS4_*re*_E** (B4), making it the least stable transition state (5.0 kcal/mol higher in energy than **TS4_*re*_Z**). These results indicated that the differentiated interaction energies are the primary determinants of the stereoselectivities.

Having proved that the cumulenolate anion generated from allenone **1** could be well captured by π-azaallyl-Pd species in Pd-catalyzed allylic alkylation reactions, we became interested in capturing the cumulenolate anion with alternative allyl-Pd species from some other precursors such as allenylethylene carbonates **6** (AECs), which was previously reported by our group for a Pd-catalyzed [3 + 3] annulation.[16a] As shown in Scheme 6, the cumulenolate anion generated in situ was readily captured by the π-oxyallyl-Pd species to give allenyl diene products (*Z*)-**8** and (*E*)-**8** under different reaction conditions. Moreover, product (*Z*)-**8** further performed an intramolecular cyclization/isomerization reaction to form [5 + 2] annulation product **7**. However, as another potential coupling product, the allene−allene coupling product was not observed in this reaction.

In this allylic alkylation reaction, we selected trisubstituted allenone **1a** and AEC **6a** as the model substrates to optimize the reaction conditions. As shown in Table 3, several ligands were screened in the presence of $Pd_2(dba)_3{\cdot}CHCl_3$ in MTBE solvent (entries 1−5). We were glad to find that when DPEPhos and XantPhos were used as ligands, [5 + 2] annulation product **7a**, which was produced in situ from allenyl diene product (*Z*)-**8a**, was obtained in 36 and 69% yields, respectively (entries 4 and 5). Further screening solvents increased the yield of product **7a** to 74% (entries 6−10). Decreasing the loading of both **6a** and the ligand, a 91% yield of **7a** was obtained (entry 12). We next tried to isolate allenyl

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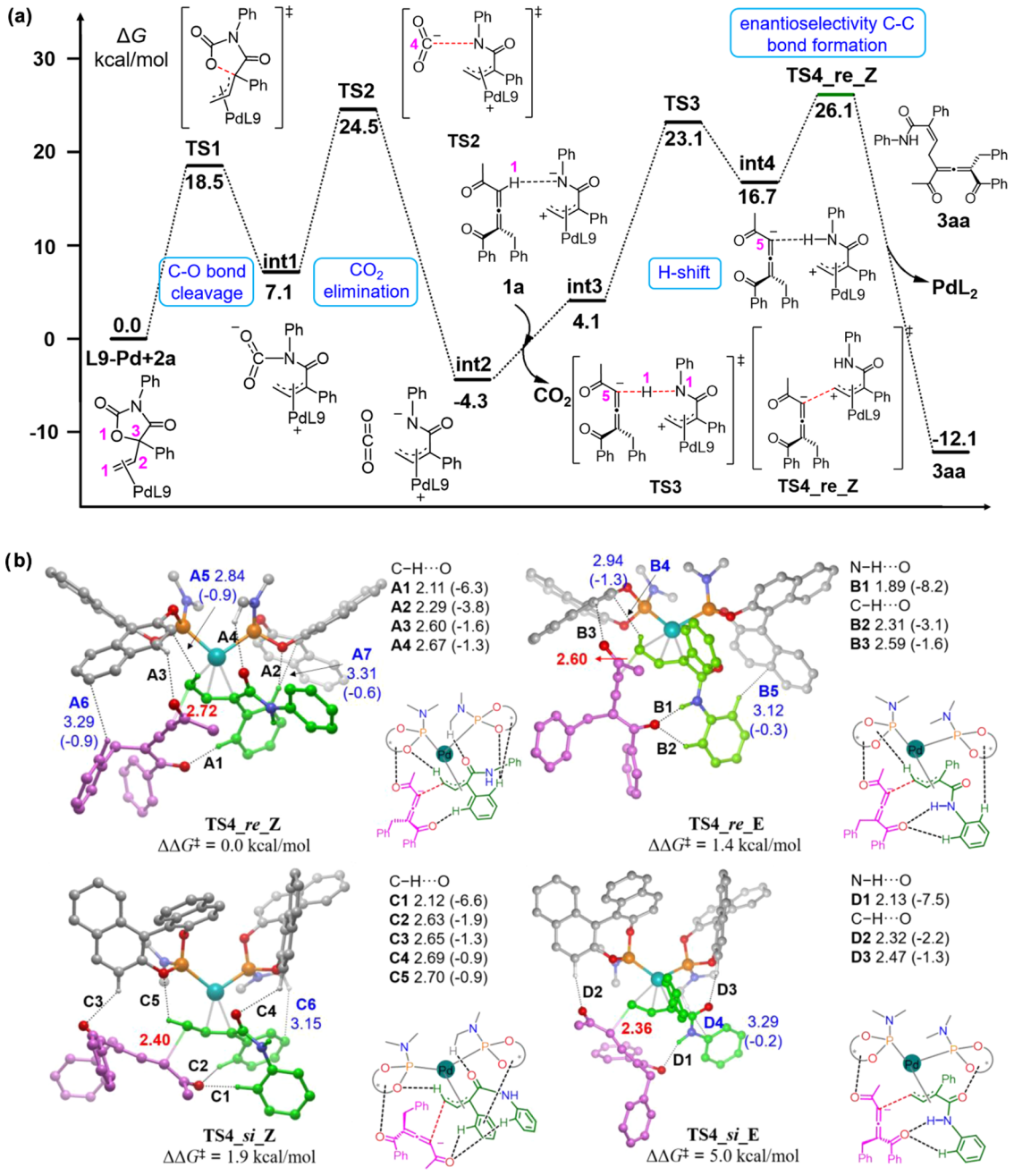


Figure 2. (a) Relative Gibbs free energy profile diagrams of the palladium-catalyzed enantioselective allylic alkylation to give **3aa** using DFT calculations (relative Gibbs energy values [Δ*G* (kcal/mol)] were calculated at the B3LYP-D3/Def2-TZVPP/SMD(THF)//B3LYP-D3/Def2-SVP/SMD (THF) level of theory). (b) Stereocontrolling C−C bond forming transition states for asymmetric allylic alkylation between allenone (**1a**) and 5-vinyloxazolidine-2,4-dione (**2a**) with catalysis by **L9**-Pd.

Table 2. Relative Gibbs Free Energies (kcal/mol) for the Enantioselective C−C Bond Forming Transition States

TS	rel ΔΔ*G* (kcal/mol)	
TS4_*re*_Z	0.0	
TS4_*re*_E	1.4	
TS4_*si*_Z	1.9	
TS4_*si*_E	5.0	
	Expt	DFT
Z/*E* ratios	11:1	13:1
ee (%)	93	94

diene products (*Z*)-**8a** and (*E*)-**8a**. By switching the Pd source from $Pd_2(dba)_3{\cdot}CHCl_3$ to $Pd(dmdba)_2$ and decreasing the reaction temperature to −10 °C, we observed allenyl diene product (*Z*)-**8a** as the sole product in the reaction mixture. However, product (*Z*)-**8a** was not able to be directly isolated, as it rapidly converted to **7a** during the purification process. Through isolating **7a** generated in situ, the yield of product (*Z*)-**8a** was determined as 51% (entry 13). Further adjustment of **6a** loading to 1.5 equiv increased the yield of (*Z*)-**8a** to 61%. Finally, under similar reaction conditions, replacing toluene with $PhCF_3$ as the solvent, allenyl diene product (*E*)-**8a** was isolated in 66% yield (entry 15). Unfortunately, screening

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Scheme 6. Pd-Catalyzed Allylic Alkylation Reaction of Allenones 1 and π-Oxyallyl-Pd Precursor 6

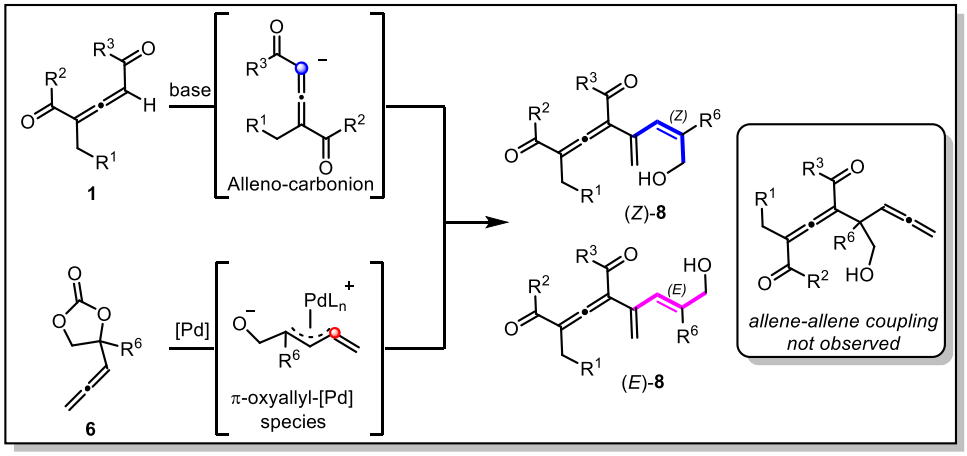
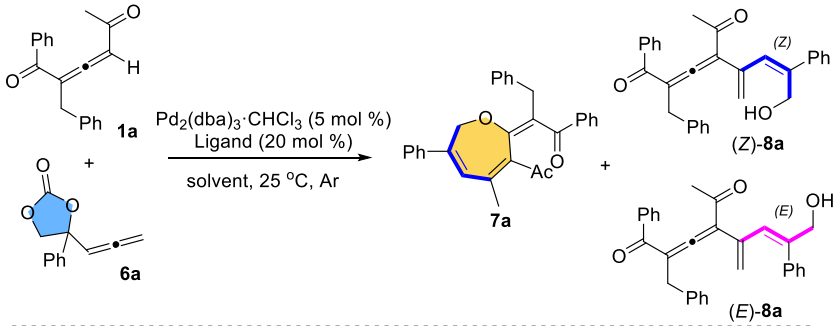


Table 3. Optimization of the Pd-Catalyzed [5 + 2] Annulation and Allylic Alkylation Reaction[a]



entry	ligand	solvent	t/h	yield (%)[b] 7a	(Z)-8a	(E)-8a
1	dppe	MTBE	24	NR[c]		
2	dppf	MTBE	24	NR[c]		
3	PPh_3	MTBE	24	NR[c]		
4	DPEPhos	MTBE	18	36		
5	XantPhos	MTBE	12	69		trace
6	XantPhos	DCM	4	38		
7	XantPhos	THF	4	48		
8	XantPhos	Et_2O	18	NR[c]		
9	XantPhos	toluene	1	74		
10	XantPhos	*m*-xylene	5	52		trace
11[d]	XantPhos	toluene	1	78		
12[d,e]	XantPhos	toluene	1	91		
13[f,g]	XantPhos	toluene	10		51	trace
14[e,f,g]	XantPhos	toluene	18		61	trace
15[e,f,g]	XantPhos	$PhCF_3$	26		trace	66

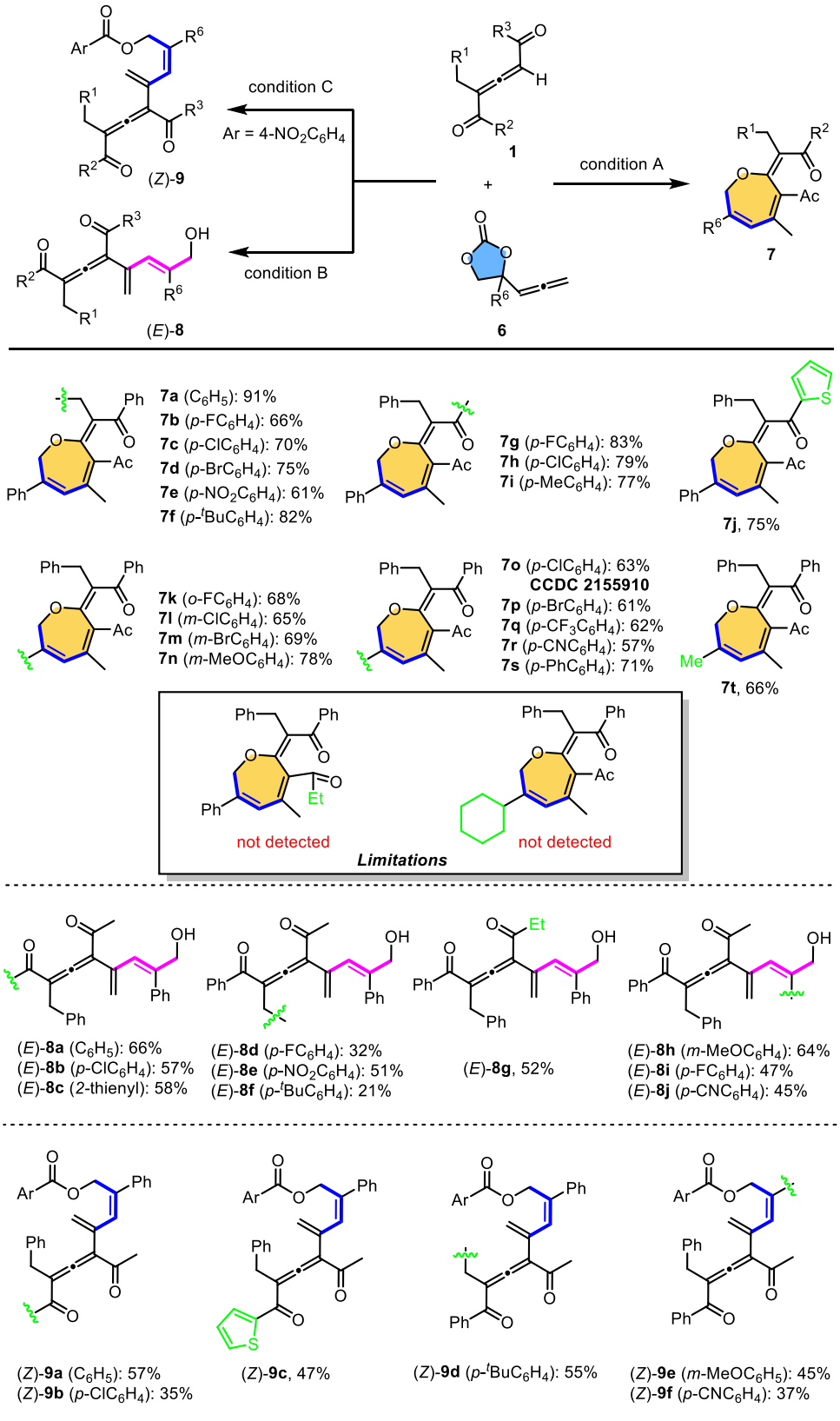
[a]All reactions of **1a** (0.10 mmol), **6a** (0.20 mmol), $Pd_2(dba)_3$·$CHCl_3$ (5 mol %), and a ligand (20 mol %) were performed in 1 mL of solvent at 25 °C. [b]Isolated yield. [c]No reaction. [d]15 mol % ligand was used. [e]0.15 mmol of **6a** was used. [f]$Pd(dmdba)_2$ was used instead of $Pd_2(dba)_3$·$CHCl_3$. [g]The reaction was performed at −10 °C.

chiral ligands failed to deliver the desired allenyl diene products (details in Table S2 in the SI).

With the optimized reaction conditions in hand, we next investigated the substrate scope of the Pd-catalyzed [5 + 2] annulation and allylic alkylation reaction (Scheme 7). First, a variety of trisubstituted allenones **1** and AECs **6** were subjected to the [5 + 2] annulation under condition A. The reactions were completed within 1 h in most cases. Various aromatic and heteroaromatic rings on allenones **1** were well tolerated in the reaction, regardless of the position or electronic properties of

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Scheme 7. Substrate Scope of Pd-Catalyzed [5 + 2] Annulation and Allylic Alkylation Reaction[a]



[a]Condition A: reactions were performed with **1** (0.10 mmol), **6** (0.15 mmol), $Pd_2dba_3{\cdot}CHCl_3$ (5 mol %), and XantPhos (15 mol %) in 1 mL of toluene at 25 °C for 0.75−8 h. Condition B: reactions were performed with **1** (0.10 mmol), **6** (0.15 mmol), $Pd(dmdba)_2$ (5 mol %), and XantPhos (20 mol %) in 1 mL of $PhCF_3$ at −10 °C for 17−34 h. Condition C: reactions were performed with **1** (0.10 mmol), **6** (0.15 mmol), $Pd(dmdba)_2$ (5 mol %), and XantPhos (20 mol %) in 1 mL of toluene at −10 °C for 18−36 h, and then PPh_3 (1.1 equiv), DEAD (1.1 equiv), and 4-nitrobenzoic acid (1.5 equiv) were added. The mixture was further stirred at −10 °C for 2−7 h before purification. All yields are isolated yields.

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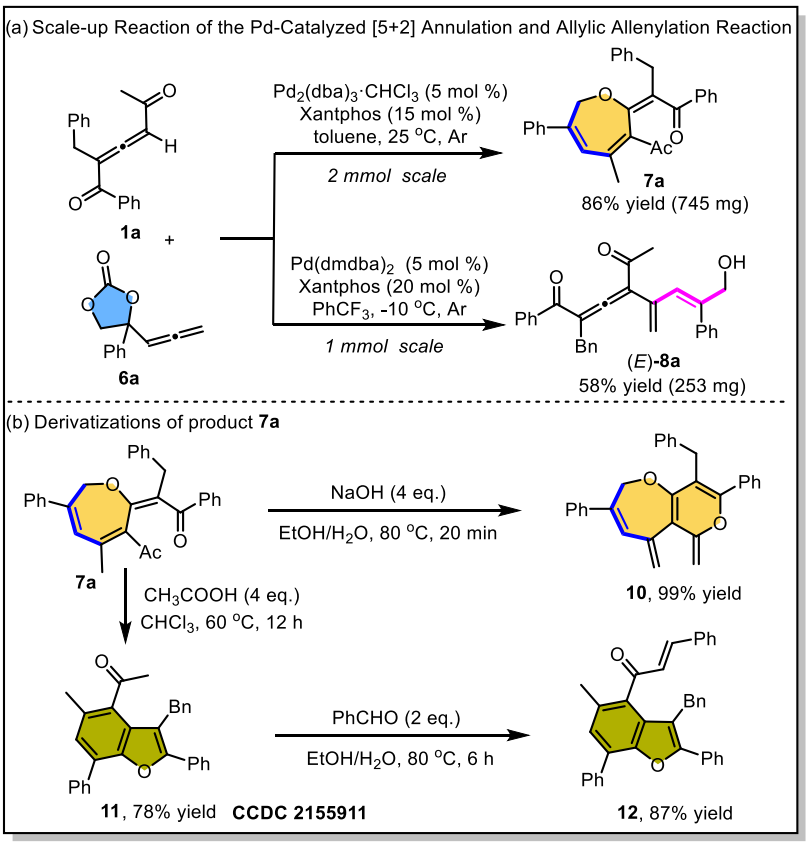
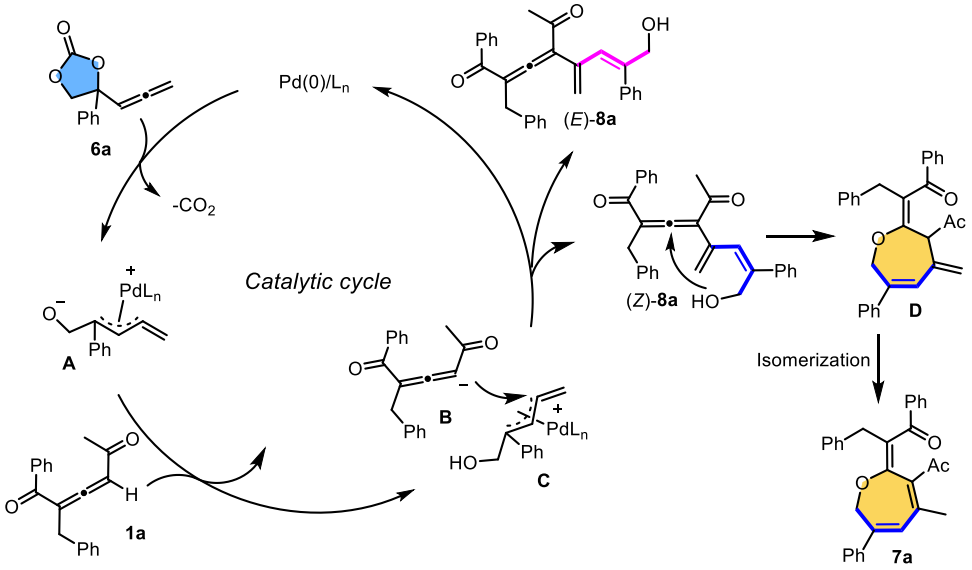


Figure 3. (a) Scale-up reaction of the Pd-catalyzed [5 + 2] annulation and allylic alkylation reaction. (b) Derivatizations of product **7a**.

Scheme 8. Plausible Reaction Mechanism for the Pd-Catalyzed Allylic Alkylation and [5 + 2] Annulation



their substituents, delivering the desired products (**7a**−**7j**) in moderate to excellent yields. It was worth noting that replacing the acetyl group with the propionyl group of allenone completely blocked the expected [5 + 2] annulation process, and only a certain amount of (*Z*)-configured allenyl diene product was detected instead of the annulation product. This result revealed that the appropriate electronic property of the acyl group is key to allowing the annulation process. Similar to the behavior of allenone **1**, AECs **6** with F, Cl, Br, MeO, CF_3, CN, and phenyl substituents on the phenyl ring all performed well to deliver corresponding products (**7k**−**7s**) in satisfactory yields. To our delight, AEC **6t** with methyl as an R^6 substituent was also a suitable substrate, producing product **7t** in 66% yield, albeit in prolonged reaction time (8 h). Nevertheless,

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when the methyl substituent was switched to cyclohexyl as R^6, no corresponding product was observed, probably due to the increased steric hindrance and lower reactivity. The structure of product **7o** was determined by X-ray crystallographic analysis.[17]

Subsequently, the scope of Pd-catalyzed allylic alkylation was examined under condition B (Scheme 7). An array of allenones with variations on R^1, R^2, and R^3 reacted well with AEC **6a** to afford (*E*)-configured allenyl diene products (*E*)-**8a**−**8g** in moderate yields. The allenones with *p*-F and *p*-^{t}Bu phenyl as R^1 delivered allenyl diene products (*E*)-**8d** and (*E*)-**8f** in lower yields (32 and 21%, respectively). Three AECs with substituted phenyl groups were also compatible substrates, generating the corresponding products **8h**−**8j** in acceptable yields. Several allenones and AEC substrates were used to evaluate the scope of Pd-catalyzed allylic alkylation for the synthesis of (*Z*)-configured allenyl diene products. In order to get a stable (*Z*)-allenyl diene product, the reaction mixture was treated with 4-nitrobenzoic acid under typical Mitsunobu coupling conditions (one-pot reaction, condition C), and coupling products (*Z*)-**9a**−**9f** were obtained in 35−57% yields, suggesting that the reaction had good tolerance of different substituents on aromatic rings of substrates.

To demonstrate the synthetic potential of the Pd-catalyzed [5 + 2] annulation and allylic alkylation reaction, 2 and 1 mmol scale-up experiments were performed, respectively, to give products **7a** and (*E*)-**8a** in 86 and 58% yields (Figure 3a). Further treatment of product **7a** with NaOH in a hot mixture of ethanol and water enabled an intramolecular dehydration annulation to produce bicyclic product **10** in 99% yield. Moreover, dropwise addition of acetic acid into the chloroform solution of **7a** led to an interesting ring contraction/aromatization reaction, giving benzofuran product **11** in 78% yield, which further underwent a typical aldol reaction with benzaldehyde to give enone product **12** in 87% yield (Figure 3b). The structure of product **11** has been unambiguously confirmed through X-ray crystallographic analysis.[17]

A plausible reaction mechanism was proposed in Scheme 8 for Pd-catalyzed [5 + 2] annulation and allylic alkylation. Under the catalysis of the [Pd] catalyst, AEC **6a** underwent ring-opening with the release of CO_2 to afford π-allyl intermediate **A**, which deprotonated the acidic C(sp^2)−H of trisubstituted allenone **1a** to generate cumulenolate anion **B** and intermediate **C**. Subsequent allylic alkylation coupling of intermediates **B** and **C** led to the formation of tetrasubstituted allenyl diene product (*Z*)-**8a** or (*E*)-**8a** and regeneration of the Pd(0) catalyst under specific reaction conditions. Besides, the unstable and highly reactive product (*Z*)-**8a** tended to further perform intramolecular nucleophilic addition to give intermediate **D**, which then underwent isomerization to form a more stable cyclic product (**7a**).

CONCLUSIONS

In summary, we achieved palladium-catalyzed allenylic C-(sp^2)−H allylation reactions for the first time. In the first part, a series of trisubstituted allenones and 5-vinyloxazolidine-2,4-diones underwent an asymmetric allylation reaction under mild reaction conditions to afford various axially chiral tetrasubstituted allenones in satisfactory yields with moderate to excellent stereoselectivity (up to 96% ee and up to 20:1 *Z*/*E*). Mechanistic investigation and DFT calculations indicated that appropriate hydrogen-bonding interaction between the acidic C(sp^2)−H of allene and the nitrogen anion of chiral π-allyl-Pd species dominated the successful stereocontrol of the allylation reaction. In the other part, we developed Pd-catalyzed allylic alkylation of allenones and allenylethylene carbonates, providing divergent access to (*E*)- and (*Z*)-configured allenyl diene products. Moreover, the subsequent sequential intramolecular cyclization/isomerization process enabled a [5 + 2] annulation reaction to give an array of dihydrooxepine derivatives. The current reactions described here revealed the high potential of allenylic nucleophiles to be applied in more transition metal-catalyzed allylation reactions, which offers more fascinating opportunities in the synthesis of functionalized allenes.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c01460.

Experimental procedures; characterization data; NMR spectra for substrates and products; HPLC chromatograms for racemic and chiral allenes; and crystallographic data (PDF)

Accession Codes

CCDC 2260721, 2290004, 2306683, 2155910, and 2155911 contain supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: + 44 1223 336033.

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‖W.W. and Y.H. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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